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MULTIMEDIA DATABASE

Cross-Modal Implicit Relation Reasoning and Aligning for Text-to-Image Person Retrieval

IRRA • CVPR 2023

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ONE LOVE. ONE FUTURE.

I. Introduction — From Description to Top-k

TEXT QUERY

**“A man
wearing a
white-and-gray
striped shirt,
green pants,
and shoes.”**

INPUT
Natural-language
description

OUTPUT
Ranked image
list

Baseline



IRRA

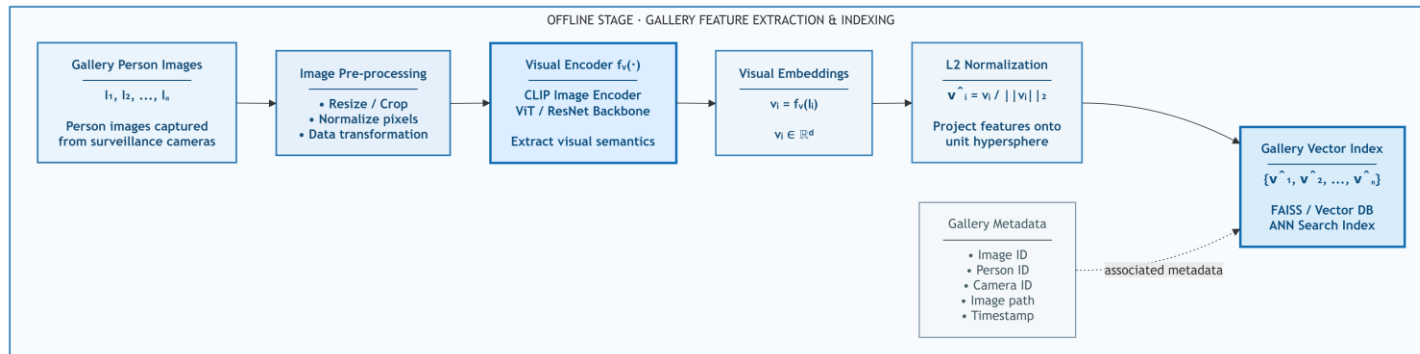


Cross-modal retrieval

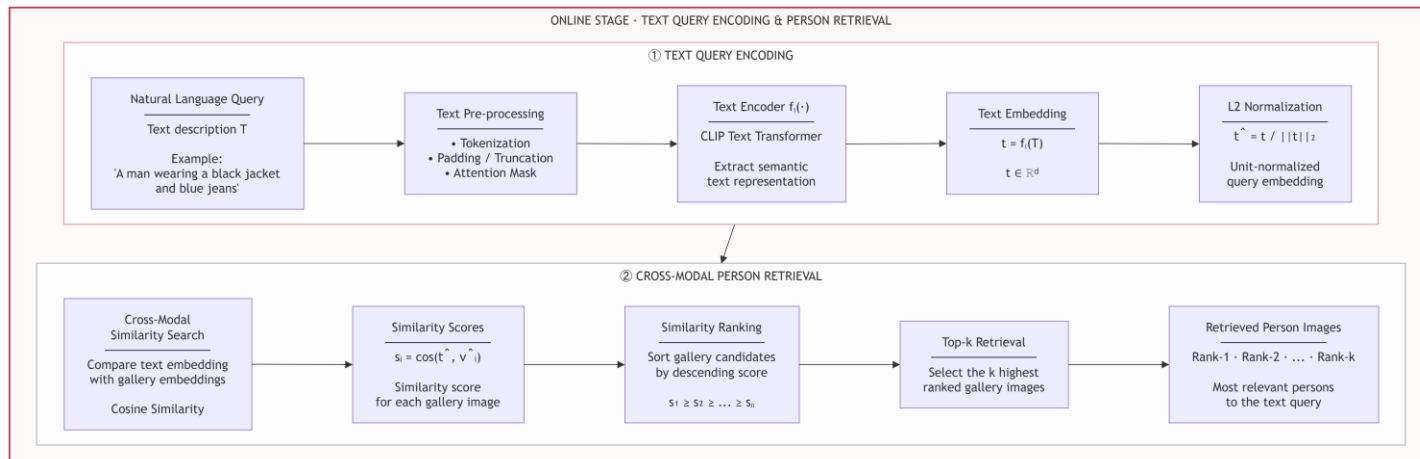
query = text → gallery = images → retrieve Top-k

I. Introduction — Multimedia Database Perspective

OFFLINE • GALLERY INDEXING



ONLINE • QUERY PROCESSING



I. Introduction — Why Is the Task Difficult?

1

Large intra-identity variation

Viewpoint • pose • lighting • occlusion

2

Modality heterogeneity

Pixels/patches ↔ tokens and word order

3

Subtle discriminative details

White/black bag • striped/plain shirt • shoes

4

Language ambiguity

Arbitrary order • missing attributes • synonyms

Learn fine-grained local cues while retaining efficient global retrieval.

I. Background — Three Matching Paradigms

GLOBAL

One image vector ↔ one text vector

Fast, but misses fine details

EXPLICIT LOCAL

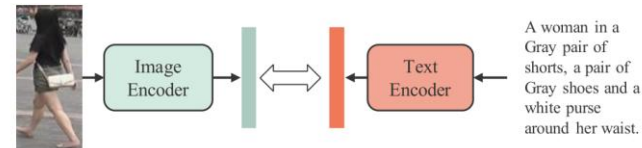
Body parts ↔ textual phrases

Noisy priors; multiple vectors to store and compare

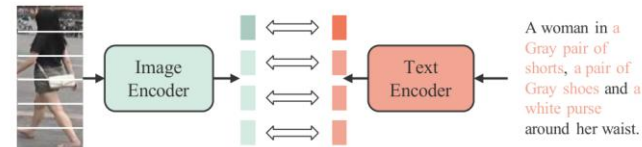
IRRA

Implicit local learning during training

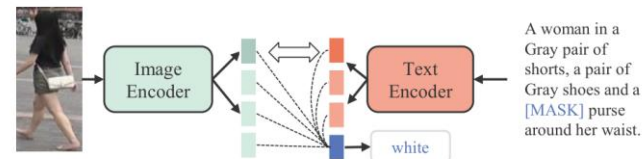
Global embeddings at inference



(a) Early global matching paradigm



(b) Existing explicit local matching paradigm



(c) Our implicit relation reasoning aided matching paradigm



I. Background — Dual-Stream Models and CLIP

SINGLE-STREAM

[image tokens + text tokens]



JOINT TRANSFORMER

- + Strong cross-modal interaction
- Recomputed for every image–text pair

DUAL-STREAM • CLIP

IMAGE
ENCODER

TEXT
ENCODER

joint embedding
space

- + Precomputed gallery embeddings
- + Image–text pre-training
- Limited fine-grained interaction

IRRA = full CLIP + interaction only during TRAINING

II. Proposed Method — Research Question

How can fine-grained image–word relations improve global embeddings
without part-level supervision or additional retrieval cost?

1

IRR / MLM

Implicitly learns fine-grained image–text relations

2

SDM

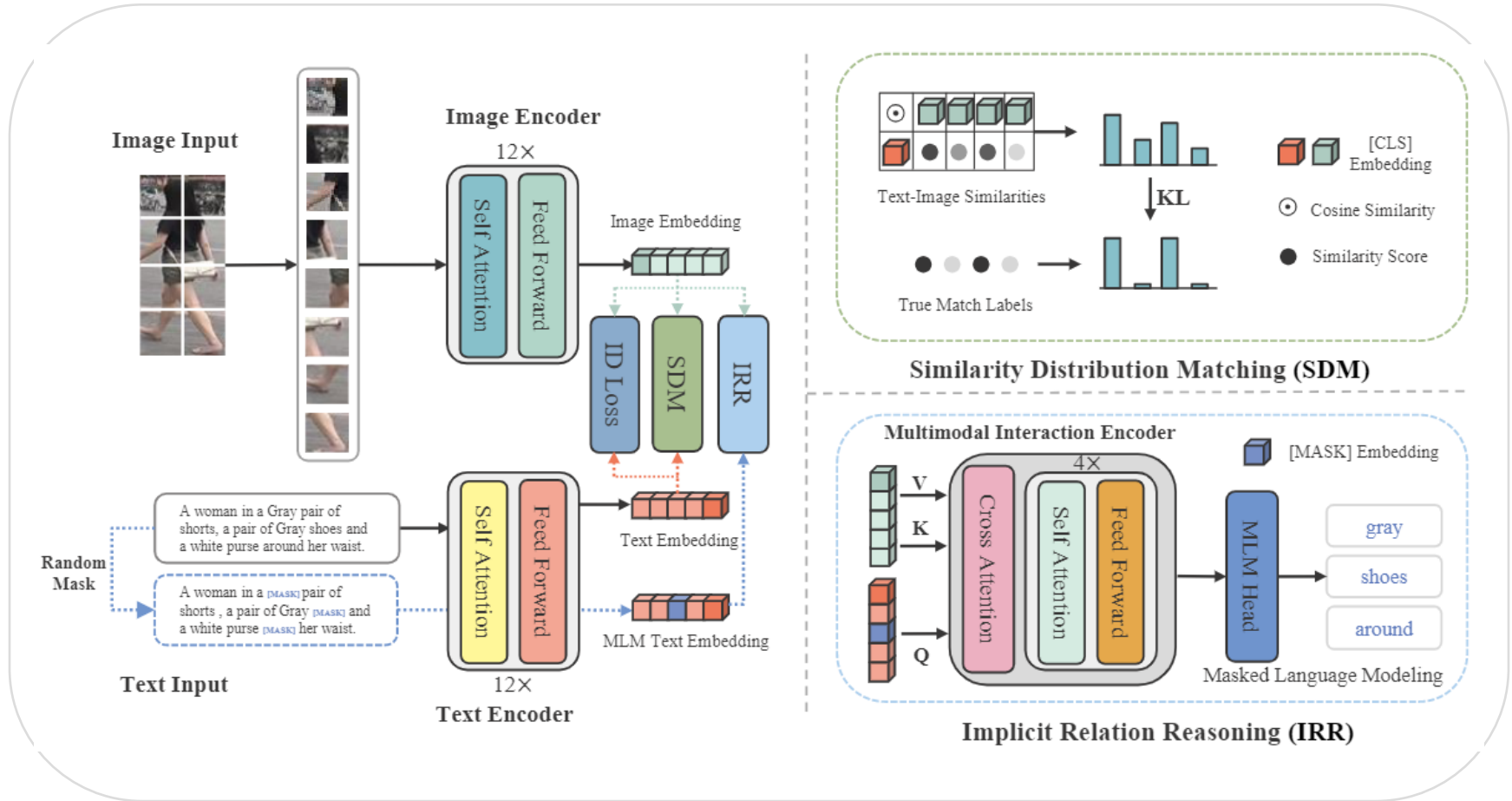
Matches similarity distributions to label distributions

3

FULL CLIP

Fine-tunes both image and text encoders

II. Proposed Method — IRRA Overview



1 • CLIP DUAL ENCODER

2 • SDM + ID

3 • IRR / MLM

Dashed modules: removed at inference

II. Proposed Method — CLIP Dual Encoder

IMAGE ENCODER • ViT-B/16

384×128

patch 16×16

$24 \times 8 = 192$ visual tokens



$[\text{CLS}] \rightarrow f_v$

12 Transformer blocks

TEXT ENCODER • Transformer

BPE vocabulary

49,152

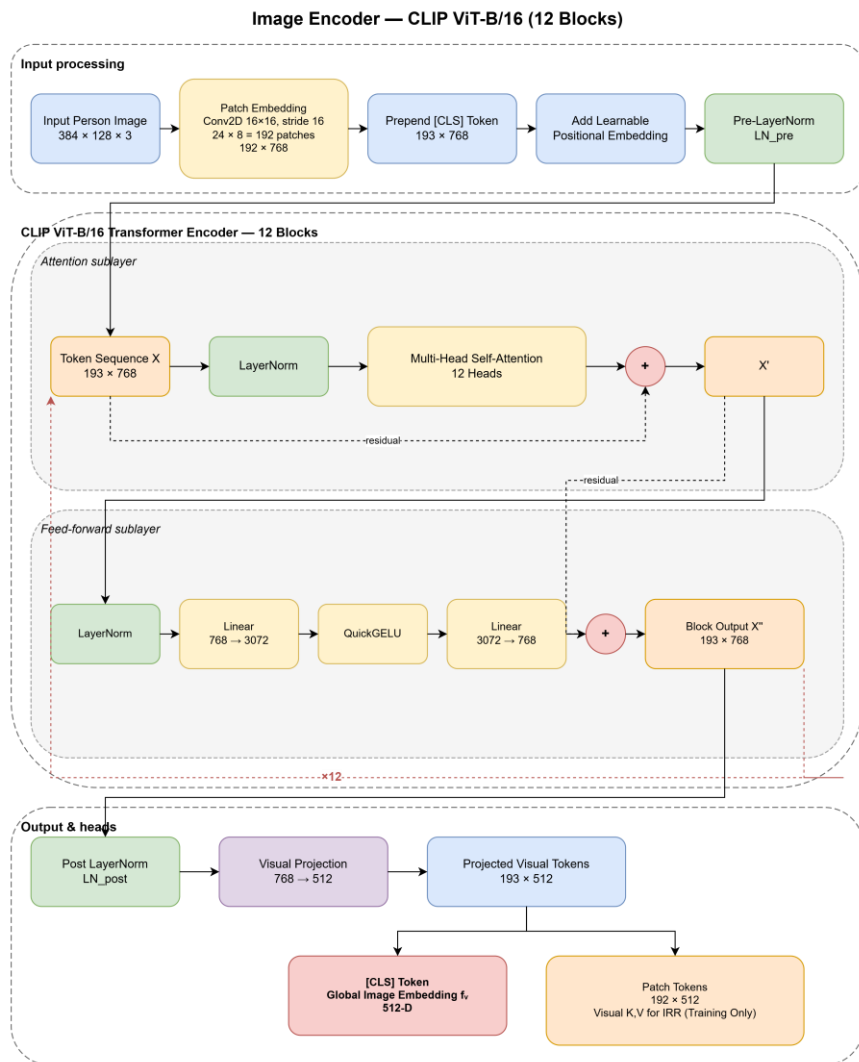
$[\text{SOS}] \text{ token}_1 \dots \text{token}_{77} [\text{EOS}]$



$[\text{EOS}] \rightarrow f_t$

12 Transformer blocks

II. Proposed Method — Inside the ViT Image Encoder



ViT-B/16 • 12 BLOCKS

1 Patch embedding
+ position

2 Pre-LN multi-head
self-attention

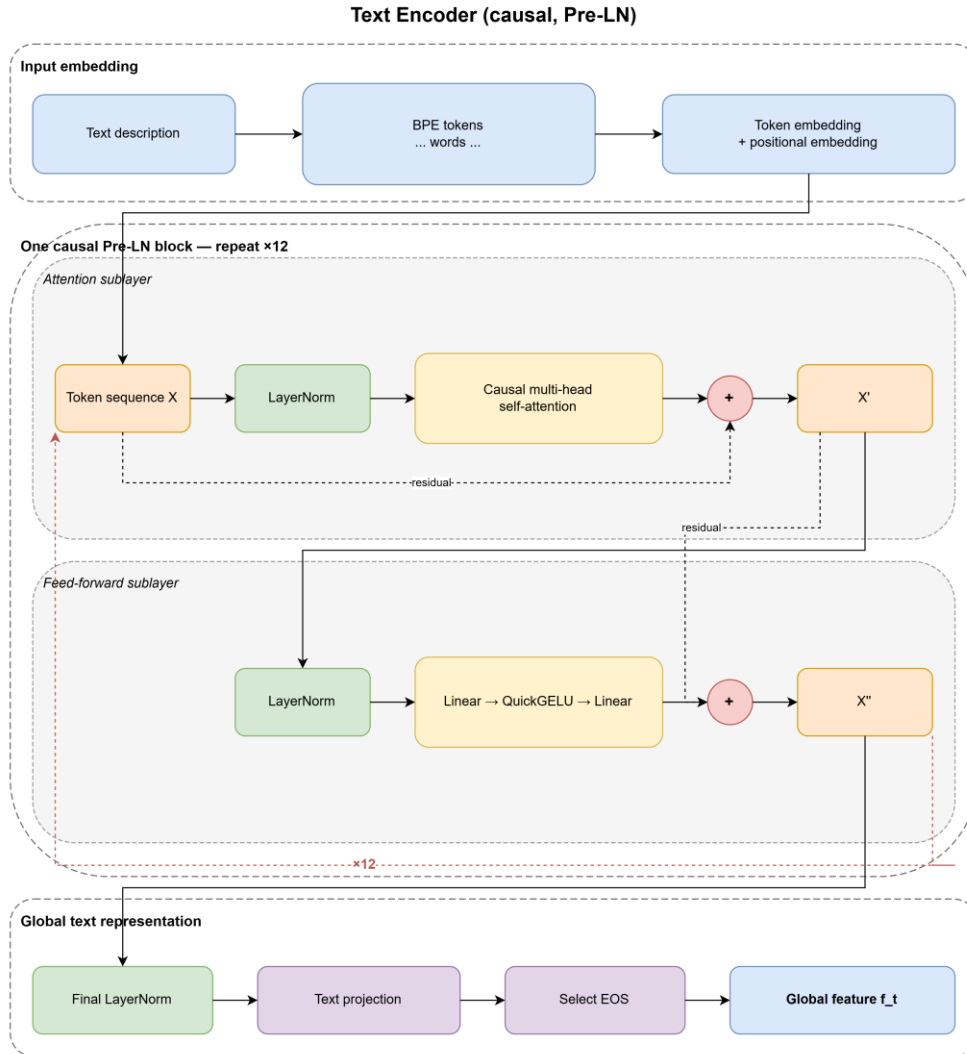
3 Residual
connection

4 Pre-LN MLP: 768
 $\rightarrow 3072 \rightarrow 768$

5 [CLS] collects
global visual context

$$\begin{aligned}x &\leftarrow x + \text{MSA}(\text{LN}(x)) \\x &\leftarrow x + \text{MLP}(\text{LN}(x))\end{aligned}$$

II. Proposed Method — Inside the CLIP Text Encoder



TEXT • 12 BLOCKS

1 BPE token + position

2 Causal self-attention

3 Same Pre-LN residual block

4 [EOS] summarizes the prefix

5 Linear projection → f_t

token i attends only to positions $\leq i$

II. Proposed Method — IRR through MLM

ORIGINAL TEXT

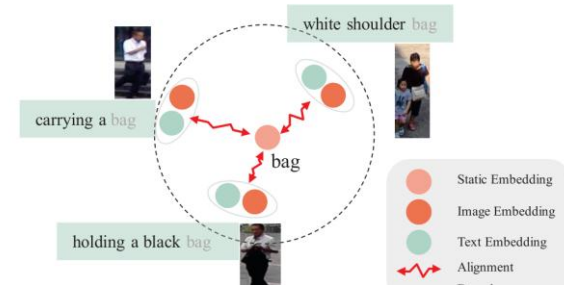
... gray shoes and a white purse around her waist.

MASKED TEXT

... gray [MASK] and a [MASK] purse [MASK] her waist.

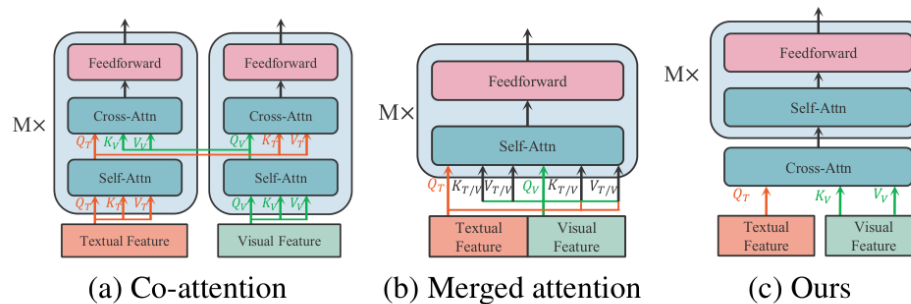
Predictions: shoes • white • around

Select 15% of tokens
80% [MASK] • 10% random • 10%
unchanged



$$\mathcal{L}_{\text{MLM}} = -\frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \log p_{\theta}(w_i | \hat{T}, I)$$

II. Proposed Method — Multimodal Interaction



Q masked text tokens

K, V visual tokens

$$\text{MCA}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right) V$$

One-way cross-attention $\rightarrow$ 4 Transformer blocks
hidden size = 512 • 8 heads

II. Proposed Method — Similarity Distribution Matching

For a mini-batch of N pairs: $\text{sim}(\cdot, \cdot)$ = cosine similarity; $y_{ij} = 1$ iff the identities match

1 • PREDICTION

$$p^i \rightarrow^t_{i,j} = \frac{\exp(\text{sim}(f^v_i, f^t_j)/\tau)}{\sum_{k=1}^N \exp(\text{sim}(f^v_i, f^t_k)/\tau)}$$

2 • TARGET

$$q^i \rightarrow^t_{i,j} = \frac{y_{i,j}}{\sum_{k=1}^N y_{i,k}}$$

3 • LOSS

$$\mathcal{L}_{\text{SDM}} = KL(p^i \rightarrow^t \parallel q^i \rightarrow^t) + KL(p^t \rightarrow^i \parallel q^t \rightarrow^i)$$

II. Proposed Method — Identity Classification Loss

$$\mathcal{L}_{\text{ID}} = \frac{1}{2} [CE(Wf^v, y) + CE(Wf^t, y)]$$

INPUT

Global image feature f^v
Global text feature f^t
One shared classifier W

LEARNS

Identity-discriminative
global clusters around
shared class prototypes

BACKPROP

Updates both CLIP
encoders
and classifier W
Classifier removed at
test

**Auxiliary objective: identity classification complements—but does not replace—
cross-modal alignment.**

II. Proposed Method — Training vs. Inference

$$L = L_{IRR} + L_{SDM} + L_{ID}$$

TRAIN

- CLIP dual encoder
- + IRR / interaction encoder
- + MLM head
- + SDM and ID classifier
- → **end-to-end local interaction learning**

INFERENCE

- Only the two encoders remain
- Precomputed gallery embeddings
- One cosine score per image–text pair
- No interaction module required
- → **efficient global retrieval**

Fine-grained learning → efficient global retrieval

III. Experiments — Three Benchmarks

Dataset	Identities	Images	Descriptions	Key property
CUHK-PEDES	13,003	40,206	80,412	2 descriptions/image
ICFG-PEDES	4,102	54,522	54,522	1 description/image
RSTPReid	4,101	20,505	41,010	15 cameras; 5 images/ID

IDENTITY-DISJOINT SPLIT

No identity appears in both training and test sets

CUHK: first benchmark

ICFG: large test gallery

RSTP: multi-camera

Test = generalization to unseen identities

III. Experiments — Metrics and Training Setup

RANK-k

At least one correct image appears in Top-k

mAP

Ranking quality across all relevant images

mINP

Position of the hardest / last relevant image

IMPLEMENTATION

- CLIP ViT-B/16
- Image size: 384×128
- Text length ≤ 77 tokens
- Adam • 60 epochs
- Learning rate: $1e-5$
- Warm-up: 5 epochs
- New modules: $5e-5$
- $\tau = 0.02$ • RTX 3090

Higher is better for all metrics

III. Results — State-of-the-Art Comparison (2023)

Method	Type	Ref	Image Enc.	Text Enc.	Rank-1	Rank-5	Rank-10	mAP	mINP
CMPM/C [53]	L	ECCV18	RN50	LSTM	49.37	-	79.27	-	-
TIMAM [36]	G	ICCV19	RN101	BERT	54.51	77.56	79.27	-	-
ViTAA [46]	L	ECCV20	RN50	LSTM	54.92	75.18	82.90	51.60	-
NAFS [12]	L	arXiv21	RN50	BERT	59.36	79.13	86.00	54.07	-
DSSL [55]	L	MM21	RN50	BERT	59.98	80.41	87.56	-	-
SSAN [7]	L	arXiv21	RN50	LSTM	61.37	80.15	86.73	-	-
LapsCore [49]	L	ICCV21	RN50	BERT	63.40	-	87.80	-	-
ISANet [51]	L	arXiv22	RN50	LSTM	63.92	82.15	87.69	-	-
LBUL [48]	L	MM22	RN50	BERT	64.04	82.66	87.22	-	-
Han et al. [13]	G	BMVC21	CLIP-RN101	CLIP-Xformer	64.08	81.73	88.19	60.08	-
SAF [28]	L	ICASSP22	ViT-Base	BERT	64.13	82.62	88.40	-	-
TIPCB [5]	L	Neuro22	RN50	BERT	64.26	83.19	89.10	-	-
CAIBC [47]	L	MM22	RN50	BERT	64.43	82.87	88.37	-	-
AXM-Net [10]	L	MM22	RN50	BERT	64.44	80.52	86.77	58.73	-
LGUR [38]	L	MM22	DeiT-Small	BERT	65.25	83.12	89.00	-	-
IVT [39]	G	ECCVW22	ViT-Base	BERT	65.59	83.11	89.21	-	-
CFine [50]	L	arXiv22	CLIP-ViT	BERT	69.57	85.93	91.15	-	-
Baseline (CLIP-RN50)	G	-	CLIP-RN50	CLIP-Xformer	57.26	78.57	85.58	50.88	34.44
Baseline (CLIP-RN101)	G	-	CLIP-RN101	CLIP-Xformer	60.27	80.88	87.88	53.93	37.54
Baseline (CLIP-ViT-B/16)	G	-	CLIP-ViT	CLIP-Xformer	68.19	<u>86.47</u>	<u>91.47</u>	<u>61.12</u>	<u>44.86</u>
IRRA (Ours)	G	-	CLIP-ViT	CLIP-Xformer	73.38	89.93	93.71	66.13	50.24

Table 1. Performance comparisons with state-of-the-art methods on CUHK-PEDES dataset. Results are ordered based on the Rank-1 accuracy. “G” and “L” in “Type” column stand for global-matching/local-matching method.

III. Results — Ablation: IRR and SDM

No.	Methods	Components			CUHK-PEDES			ICFG-PEDES			RSTPReid		
		SDM	$\mathcal{L}_{id}$	IRR	Rank-1	Rank-5	Rank-10	Rank-1	Rank-5	Rank-10	Rank-1	Rank-5	Rank-10
0	Baseline				68.19	86.47	91.47	56.74	75.72	82.26	54.05	80.70	88.00
1	+ $\mathcal{L}_{cmpm}$ [53]				59.31	79.66	86.11	53.83	72.20	79.02	55.40	77.70	85.25
2	+SDM	✓			70.42	86.73	92.04	60.45	77.88	83.86	57.20	79.90	88.10
3	+ $\mathcal{L}_{id}$		✓		65.33	84.05	90.33	53.38	72.70	79.70	54.15	76.65	85.00
4	+IRR			✓	71.23	88.89	93.24	60.96	79.02	84.90	57.90	80.85	88.50
5	+SDM+ $\mathcal{L}_{id}$	✓	✓		70.52	87.59	92.12	61.03	78.26	83.89	58.65	80.70	87.05
6	+SDM +IRR	✓		✓	<u>72.81</u>	<u>89.31</u>	<u>93.39</u>	<u>63.27</u>	<u>80.10</u>	<u>85.77</u>	<u>59.25</u>	79.70	88.00
7	IRRA	✓	✓	✓	73.38	89.83	93.71	63.46	80.25	85.82	60.20	81.30	88.20

Table 4. Ablation study on each component of IRRA on CUHK-PEDES, ICFG-PEDES and RSTPReid.

III. Results — Efficiency and Qualitative Results

A man wearing a white and gray stripe shirt, a pair of green pants and a pair of shoes.



The man is crossing the street wearing a white shirt, black pants and carrying a black book bag.



The woman wears a teal t-shirt, black pants, and blue tennis shoes. She wears glasses and carries a red backpack.



Why does IRRA perform better?

IRR exploits subtle discriminative attributes:
clothing • colors • shoes • bags

Figure 5. Comparison of top-10 retrieved results on CUHK-PEDES between Baseline (the first row) and IRRA (the second row) for each text query. The image corresponding to query text., matched and mismatched images are marked with black, green and red rectangles, respectively.

IV. Discussion — Limitations and Critical Analysis

1

Phrase-level semantics

Token masking may distort phrase-level meaning.

2

Hard positives

ICFG mINP = 7.93: difficult positives remain far down the ranking.

3

Generalization

No cross-dataset, multilingual, or new-domain evaluation.

4

Deployment and ethics

No ANN benchmark; privacy, bias, and surveillance risks.

(1) is acknowledged by the authors; (2)–(4) are additional critical observations.

THREE KEY TAKEAWAYS

1

Local relation learning

during training; global
embeddings at inference

2

IRR + SDM

fine-grained interaction and
distribution alignment

3

SOTA at CVPR 2023

on CUHK-PEDES, ICFG-
PEDES, and RSTPReid

POTENTIAL EXTENSIONS

Phrase masking • hard-sample mining • multilingual retrieval • vector databases

THANK YOU • Q&A

[FULL NAME – STUDENT ID – GROUP]

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